

TASK:03

TITLE LAB 5 ACTIVITY 3

.MODEL SMALL

.STACK 100H

.CODE

MAIN PROC

; SET VIDEO MODE 2 (MONOCHROME)

MOV AX, 2

INT 10H

; SET DISPLAY PAGE AND CURSOR POSITION

MOV DX, 0727H

MOV AH, 2

INT 10H

; INITIALIZE LOOP COUNTER

MOV CX, 9

LOOP1:

; DISPLAY '*'

MOV AH, 2

MOV DL, '*'

INT 21H

; MOVE TO THE NEXT LINE

MOV AH, 2

MOV DL, 0AH

INT 21H

; MOVE CURSOR TO THE BEGINNING OF THE LINE

MOV AH, 2

MOV DL, 08H

INT 21H

; DECREMENT LOOP COUNTER

DEC CX

```
; REPEAT LOOP UNTIL CX BECOMES ZERO  
JNZ LOOP1
```

```
; SET DISPLAY PAGE AND CURSOR POSITION AGAIN  
MOV DX, 0727H  
MOV AH, 2  
INT 10H
```

```
; SET A NEW LOOP COUNTER  
MOV CX, 7
```

```
LOOP2:  
; DISPLAY '*'  
MOV AH, 2
```

```
MOV DL, '*'  
INT 21H
```

```
; DECREMENT LOOP COUNTER  
DEC CX
```

```
; REPEAT LOOP UNTIL CX BECOMES ZERO  
JNZ LOOP2
```

```
; SET DISPLAY PAGE AND CURSOR POSITION AGAIN  
MOV DX, 0B27H  
MOV AH, 2  
INT 10H
```

```
; SET A NEW LOOP COUNTER  
MOV CX, 6
```

```
LOOP3:  
; DISPLAY '*'  
MOV AH, 2  
MOV DL, '*'
```

INT 21H

; DECREMENT LOOP COUNTER
DEC CX

; REPEAT LOOP UNTIL CX BECOMES ZERO
JNZ LOOP3

; SET DISPLAY PAGE AND CURSOR POSITION
MOV DX, 072DH
MOV AH, 2
INT 10H

; INITIALIZE LOOP COUNTER
MOV CX, 9

LOOP4:
; DISPLAY '*'
MOV AH, 2
MOV DL, '*'
INT 21H

; MOVE TO THE NEXT LINE
MOV AH, 2
MOV DL, 0AH
INT 21H

; MOVE CURSOR TO THE BEGINNING OF THE LINE
MOV AH, 2
MOV DL, 08H
INT 21H

; DECREMENT LOOP COUNTER
DEC CX

; REPEAT LOOP UNTIL CX BECOMES ZERO
JNZ LOOP1

```

; SET DISPLAY PAGE AND CURSOR POSITION AGAIN
MOV DX, 072DH
MOV AH, 2
INT 10H

; SET A NEW LOOP COUNTER
MOV CX, 7

; EXIT TO DOS
MOV AH, 4CH
INT 21H

MAIN ENDP
END MAIN

```



1. Fill out the missing values, using target address calculation, in the following table.

IP	OPCODE	CODE
E06C	733F	JNC CODE1
E06E		MOV AX, 3456H

E072	7139	JNO CODE1
E074		MOV CX, 100H
E06C	8EBD	C8: MOV DS, AX
E0A7	EBE3	JMP C8
E0A9		CMP AX
E0AD	F4	CODE1: HLT